

Statistical Models for Discrete Outcomes

Session 1 - Introduction to the module, review of linear regression and introduction to logistic regression

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Module overview

Module objectives

To equip students with the skills to:

- i) understand the principles underlying statistical models for binary, count and categorical outcomes, and
- ii) be able to analyse data using such models.

By the end of this study module you should be able to

- ▶ Understand the theoretical basis of Generalized Linear Models (GLMs).
- ▶ Use appropriate models for the analysis of binary, categorical and count data.
- ▶ Present results clearly and accurately in a structured report.
- ▶ Understand the concepts of confounding and associations versus causation, and how to apply these in modelling choices and interpretation.
- ▶ Have an appreciation of different analysis strategies.

Materials

- ▶ 15 lectures given by Jonathan Bartlett (8), Ruth Keogh (5), Dea Hazewinkel (2).
- ▶ 15 live practical sessions.
- ▶ The course manual (on Moodle) has the most detailed material. Lecture slides should be seen as secondary.
- ▶ There is access to papers and books through moodle.
- ▶ We will use R primarily, but Stata practical sheets and do files for the practicals will also be available.

Assessment

- ▶ There is an assignment session on 3rd February.
- ▶ The assignment will be handed out the previous week.
- ▶ The deadline for submission is noon on 11th February.
- ▶ The assessment will involve preparing a report of a data analysis.
- ▶ As well as the report, an R script or Stata do file will also need to be submitted.

Review of linear regression

Aims and objectives

Linear regression is a tool for modelling continuous outcomes. Generalized linear models (GLMs) are a larger class of models for a wider range of outcomes, including binary and count.

In this session we will:

- ▶ revise the linear regression model,
- ▶ introduce the exponential family of distributions,
- ▶ define the components of a GLM,
- ▶ introduce one such GLM, the logistic regression model for Bernoulli outcomes,
- ▶ learn the basic commands to fit GLMs in R.

Simple linear regression

- ▶ For measurements (y_i, x_i) made on each of n individuals the simple linear regression model of y on x can be written as

$$y_i = \alpha + \beta x_i + \epsilon_i.$$

- ▶ The OLS estimators of α and β can be derived as the values which minimize the squared differences between observations and predictions.
- ▶ For inference, we often make the stronger assumptions that $\epsilon_i \sim N(0, \sigma^2)$ and are independent.
- ▶ Alternatively we can say that the random variable Y_i has the conditional distribution $Y_i \mid x_i \sim N(\alpha + \beta x_i, \sigma^2)$.
- ▶ With this assumption, we have a parametric model and can estimate α and β using maximum likelihood.

Example: Morphine and mental activity trial

- ▶ A randomized trial comparing the effects of morphine and placebo on mental activity.
- ▶ 24 individuals randomized to each group.
- ▶ Mental activity assessed by the sum of scores on 7 questionnaire items, with a high score indicating high activity.
- ▶ Mental activity scores recorded 2 hours after injection of the drug.

Placebo						Morphine					
7	1	10	10	6	2	4	2	14	0	2	0
7	5	6	6	3	8	6	0	1	6	5	3
0	11	10	0	8	6	0	10	0	0	1	2
7	1	8	5	1	5	9	0	2	7	2	12

Example: Morphine and mental activity trial

```
library(haven)
mental <- read_dta("mental.dta")
summary(mental)
```

	treat		prement		mentact
Min.	:1	Min.	: 0.000	Min.	: 0.000
1st Qu.:	:1	1st Qu.:	: 3.750	1st Qu.:	: 0.000
Median	:2	Median	: 6.000	Median	: 2.000
Mean	:2	Mean	: 5.875	Mean	: 3.792
3rd Qu.:	:3	3rd Qu.:	: 8.000	3rd Qu.:	: 6.250
Max.	:3	Max.	:14.000	Max.	:14.000

The variable `treat` takes the value 1 in the placebo group, 2 in the morphine group, and 3 in the heroin group. For now we analyse only those in the placebo and morphine groups

Notice here that R has `treat` stored as a numeric variable (since the mean and medians have been reported).

R analysis of the morphine and mental activity trial

```
mentactmod1 <- lm(mentact~factor(treat), data=subset(mental, treat<3))  
summary(mentactmod1)
```

Call:

```
lm(formula = mentact ~ factor(treat), data = subset(mental, treat <  
3))
```

Residuals:

Min	1Q	Median	3Q	Max
-5.5417	-3.5729	-0.5417	2.3646	10.3333

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.5417	0.7669	7.226	4.18e-09 ***
factor(treat)2	-1.8750	1.0846	-1.729	0.0905 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.757 on 46 degrees of freedom

Multiple R-squared: 0.06101, Adjusted R-squared: 0.0406

F-statistic: 2.989 on 1 and 46 DF, p-value: 0.09055

R analysis of the morphine and mental activity trial

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.541667	0.7668941	7.226118	4.177344e-09
factor(treat)2	-1.875000	1.0845520	-1.728824	9.054775e-02

1. Provide an interpretation of both estimated regression coefficients.
2. What is the estimated residual variance?

R analysis of the morphine and mental activity trial

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.541667	0.7668941	7.226118	4.177344e-09
factor(treat)2	-1.875000	1.0845520	-1.728824	9.054775e-02

5.541667: Estimated mean mental activity score on placebo.

-1.875: Estimated difference in means between the morphine and placebo groups (morphine - placebo).

14.115 (3.757^2): Estimated residual variance.

Multiple linear regression

- ▶ Now we have two predictor variables (z_i as well as x_i).
- ▶ A multiple linear regression model can be written:

$$y_i = \alpha + \beta x_i + \gamma z_i + \epsilon_i.$$

- ▶ β here represents the effect of a 1 unit increase in x on y holding z **constant**.
- ▶ The model implies that this effect is the same whatever this constant level of z is.
- ▶ This type of model estimates the effect of x controlling (or adjusting) for z . In epidemiological applications we might say that we are controlling for the confounding effect of z .

Interaction terms

- ▶ We can extend this model to include an interaction between x and z as follows:

$$y_i = \alpha + \beta x_i + \gamma z_i + \delta x_i z_i + \epsilon_i.$$

- ▶ Adding the interaction term changes the interpretation of β .
- ▶ It is now the effect of a 1 unit increase in x on y holding z **constant at zero**.
- ▶ If z is held constant at some other value, say k , then the effect of a 1 unit increase in x on y is $\beta + k\delta$.
- ▶ See the relevant lecture in the linear regression course for more details.

Example: Morphine and mental activity trial (continued)

Mental activity before treatment was also measured.

We now fit a model that includes this as a covariate, and the interaction with treatment:

```
mentactInter <- lm(mentact~factor(treat)*prement,  
                   data=subset(mental, treat<3))
```

Example: Morphine and mental activity trial (continued)

Call:

```
lm(formula = mentact ~ factor(treat) * prement, data = subset(mental,
  treat < 3))
```

Residuals:

	Min	1Q	Median	3Q	Max
	-7.8281	-2.0250	-0.1053	1.8176	6.1719

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.97803	1.21662	1.626	0.11112
factor(treat)2	-1.21174	1.64558	-0.736	0.46542
prement	0.59394	0.17249	3.443	0.00127 **
factor(treat)2:prement	-0.08953	0.23348	-0.383	0.70324

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.133 on 44 degrees of freedom

Multiple R-squared: 0.3753, Adjusted R-squared: 0.3327

F-statistic: 8.81 on 3 and 44 DF, p-value: 0.0001093

Example: Morphine and mental activity trial (continued)

1. Interpret each estimated regression coefficient.
2. Is R^2 guaranteed to be larger here than in the earlier model?
3. Is the estimated residual variance guaranteed to be smaller here than in the earlier model?

Example: Morphine and mental activity trial (continued)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.97803030	1.2166160	1.6258461	0.1111246
factor(treat)2	-1.21174195	1.6455801	-0.7363616	0.4654187
prement	0.59393939	0.1724872	3.4433817	0.0012723
factor(treat)2:prement	-0.08952577	0.2334819	-0.3834378	0.7032417

1.97803: Estimated mean mental activity score on placebo **when the baseline score is 0.**

-1.211742: Estimated difference in means between the morphine and placebo groups **when the baseline score is 0.**

0.5939394: Estimated increase in mental activity score per 1 unit increase in baseline score (i.e. the slope) **on placebo.**

-0.0895258: Estimated difference in slopes between the morphine and placebo groups (morphine - placebo).

Example: Morphine and mental activity trial (continued)

```
summary(mentactInter)$r.squared
```

```
[1] 0.3752569
```

R^2 **is** guaranteed to be larger here than in the earlier model.

Example: Morphine and mental activity trial (continued)

```
anova(mentactInter)
```

Analysis of Variance Table

Response: mentact

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(treat)	1	42.19	42.188	4.2969	0.04407 *
prement	1	215.85	215.851	21.9850	2.675e-05 ***
factor(treat):prement	1	1.44	1.444	0.1470	0.70324
Residuals	44	432.00	9.818		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#estimated residual variance is

```
summary(mentactInter)$sigma^2
```

```
[1] 9.81811
```

The estimated **residual sum of squares** is guaranteed to be smaller here than in the earlier model. But the estimated residual variance **is not** guaranteed to be smaller, because the residual degrees of freedom is also decreased.

Maximum likelihood estimates for linear regression

Inference for regression models can be based on the (log-) likelihood function corresponding to the conditional distribution of the outcomes given the covariates.

For simple linear regression the log-likelihood given a sample $(y_i, x_i), i = 1, \dots, n$ is

$$\ell(\alpha, \beta, \sigma^2 | \mathbf{y}, \mathbf{x}) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2.$$

Differentiating with respect to α , β , and σ^2 , we obtain the MLEs as follows.

$$\begin{aligned}\hat{\beta} &= \frac{\sum x_i y_i - n \bar{x} \bar{y}}{\sum x_i^2 - n \bar{x}^2} \\ \hat{\alpha} &= \bar{y} - \hat{\beta} \bar{x} \\ \hat{\sigma}^2 &= \frac{\sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta} x_i)^2}{n}\end{aligned}$$

Maximum likelihood estimates for simple linear regression

- ▶ For α and β , the MLEs are identical to the OLS estimators.
- ▶ For σ^2 , the MLE differs from the usual (unbiased) estimator by a factor of $\frac{n-2}{n}$.
- ▶ This is analogous to the fact the MLE of σ^2 for a $N(\mu, \sigma^2)$ model differs from the usual sample variance estimator.
- ▶ A corollary of the need to estimate the residual variance in linear regression is that test statistics follow F distributions, rather than χ^2 distributions.

Motivation for generalized linear models

- ▶ Linear regression models, e.g. $Y|x \sim N(\alpha + \beta x, \sigma^2)$, assume that Y is normally distributed given x , with constant variance.
- ▶ They also assume that the expectation of Y is a linear function of x , or more generally a linear combination of covariates multiplied by regression coefficients.
- ▶ For many outcomes these assumptions will not be appropriate.
- ▶ e.g. the outcome is whether or not a patient dies during follow-up.
- ▶ e.g. the outcome is the number of asthma attacks occurring for a patient during follow-up.

Going beyond linear models

- ▶ In these examples, using a linear model is (usually) a bad idea.
- ▶ The modelling assumptions do not hold, and we will obtain invalid inferences.
- ▶ Generalized linear models generalize the linear model to enable us to model a much wider class of dependent/outcome variables.
- ▶ This wider class of distributions is the *exponential family* of distributions.

The normal distribution

The density of a $N(\mu, \sigma^2)$ random variable Y is

$$f(y) = \sqrt{\frac{1}{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (y - \mu)^2 \right\}$$

and the logarithm of this is

$$\begin{aligned} \log\{f(y)\} &= -\frac{y^2}{2\sigma^2} + \frac{y\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \\ &= \frac{y\mu - \mu^2/2}{\sigma^2} - \left\{ \frac{y^2}{2\sigma^2} + \log(2\pi\sigma^2)/2 \right\}. \end{aligned}$$

The exponential family

The exponential family of distributions consists of those distributions which can be expressed as

$$\log\{f(y)\} = \frac{y\theta - b(\theta)}{\phi} - c(y, \phi).$$

For the normal distribution we had

$$\log\{f(y)\} = \frac{y\mu - \mu^2/2}{\sigma^2} - \left\{ \frac{y^2}{2\sigma^2} + \log(2\pi\sigma^2)/2 \right\}$$

which belongs to the exponential family with the following parameterization.

$$\theta = \mu$$

$$\phi = \sigma^2$$

$$b(\theta) = \frac{\mu^2}{2}$$

$$c(y, \phi) = \frac{y^2}{2\sigma^2} + \frac{1}{2} \log(2\pi\sigma^2)$$

Canonical and scale parameters, and canonical link function

- ▶ θ is called the **canonical** or **natural** parameter of location.
- ▶ ϕ is called the scale or dispersion parameter.
- ▶ The function $g()$ which maps the mean of Y (μ) to the canonical parameter is called the canonical link function.
- ▶ For the normal distribution, the canonical link function is just the identity: $g(\mu) = \mu$.

The binomial distribution

Now consider $Y \sim \text{Bin}(n, \pi)$. Then

$$f(y) = \Pr(Y = y) = \binom{n}{y} \pi^y (1 - \pi)^{n-y}, \quad y = 0, 1, \dots, n.$$

$$\begin{aligned} \log\{f(y)\} &= y \log(\pi) + (n - y) \log(1 - \pi) + \log \left\{ \binom{n}{y} \right\} \\ &= y \log \left(\frac{\pi}{1 - \pi} \right) + n \log(1 - \pi) + \log \left\{ \binom{n}{y} \right\}. \end{aligned}$$

So $\theta = \log \left(\frac{\pi}{1 - \pi} \right)$, $\phi = 1$, $b(\theta) = -n \log(1 - \pi)$, and

$$c(y, \phi) = -\log \left\{ \binom{n}{y} \right\}$$

Although $E(Y) = n\pi$, we usually say that the canonical link for the binomial is $g(\pi) = \log \left(\frac{\pi}{1 - \pi} \right)$. The scale factor is 1.

Properties of distributions in the the exponential family

Suppose that we have a random variable Y that follows a distribution in the exponential family, and that we have a single observation from this distribution y . The following results can be derived from likelihood theory.

1. $E(Y) = b'(\theta)$.
2. $\text{Var}(Y) = \phi b''(\theta)$.
3. $\hat{\theta}$ solves $b'(\theta) = y$.
4. Asymptotically $\text{Var}(\hat{\theta}) = \frac{\phi}{b''(\theta)}$.

Modelling

- ▶ The exponential family contains many useful distributions, and they may be suitable for the sorts of examples we talked about at the start.
- ▶ We need a modelling framework to allow us to model how Y depends on covariates $(x_1, \dots, x_p)$.
- ▶ Having specified a model, we will then need an approach for estimation, testing, and confidence interval construction.

The generalized linear model

1. *Response Distribution:* The response variables $Y_i, i = 1, \dots, n$ are assumed to be independent, arising from an exponential family distribution, with $E(Y_i) = \mu_i$.
2. *Linear Predictor:* The explanatory variables $(x_1, \dots, x_p)$ enter the model in a linear combination with unknown parameters: for the i th response we have the linear predictor:

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}.$$

3. *Link Function:* The link function relates the linear predictor η_i to the mean μ_i :

$$g(\mu_i) = \eta_i.$$

Example 1: Simple Linear regression

1. *Response Distribution*: $Y_i, i = 1, \dots, n$ are assumed to be independent, arising from a normal distribution, with mean μ_i and variance σ^2 .
2. *Linear Predictor*: There is only a single explanatory variable x :

$$\eta_i = \beta_0 + \beta_1 x_i.$$

3. *Link Function*: The link function is the **identity** function:

$$g(\mu_i) = \mu_i = \eta_i.$$

Example 2: Logistic regression for a Bernoulli outcome with a single explanatory variable

1. *Response Distribution:* $Y_i, i = 1, \dots, n$ are assumed to be independent, arising from a Bernoulli distribution, with probability of success (i.e. mean) π_i (conventionally we use π_i rather than μ_i).
2. *Linear Predictor:* There is only a single explanatory variable x :

$$\eta_i = \beta_0 + \beta_1 x_i.$$

3. *Link Function:* The link function is the **logit** ($\log\left(\frac{\pi_i}{1-\pi_i}\right)$) function:

$$g(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i.$$

The logit link function

The logit is the canonical link function for the Bernoulli distribution.

$$g(\pi_i) = \log \left(\frac{\pi_i}{1 - \pi_i} \right) = \eta_i.$$

This can be re-arranged as

$$\pi_i = \frac{e^{\eta_i}}{1 + e^{\eta_i}}.$$

This formula is sometimes termed the *expit* function.

In R there is a function `plogis` that calculates it.

Correct and incorrect model specifications

In example 1 we could (having specified that residuals are independent) specify the model as:

$$y_i = \alpha + \beta x_i + \epsilon_i : \epsilon_i \sim N(0, \sigma^2).$$

However, because Y_i only takes the values 0 and 1, we cannot write the model in example 2 as:

$$\log \left(\frac{y_i}{1 - y_i} \right) = \alpha + \beta x_i + \epsilon_i.$$

Instead, the model can be written succinctly as

$$Y_i \sim \text{Bin} \left(1, \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right).$$

Standard model specification

- ▶ It is customary when specifying a GLM to separate the description of the distribution from those for the other components (the linear predictor and the link function).
- ▶ Sometimes these latter two are combined in a single formula (omitting η_i), so we could write the model in example 2 as:

$$Y_i \sim \text{Bin}(1, \pi_i) : \log \left(\frac{\pi_i}{1 - \pi_i} \right) = \beta_0 + \beta_1 x_i,$$

$$\text{or } Y_i \sim \text{Bin}(1, \pi_i) : \pi_i = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}.$$

Which bit is linear?

The linear in generalized linear models refers to the fact that the linear predictor is a linear combination of the parameters.

For example, in polynomial regression, powers of the variables occur:

$$\eta_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_p x_i^p.$$

On the other hand

$$\eta_i = \beta_0(1 - e^{\beta_1 x_{i1}})$$

is **not** a linear predictor.

The link function

- ▶ The link function should be chosen such that the covariate effects are expected to be additive.
- ▶ Frequently we use link functions whose inverses map only to valid values of the mean parameter.
- ▶ For example, for the binomial distribution $0 < \pi < 1$, so ideally we would consider link functions whose inverses map to $(0, 1)$.
- ▶ Conversely, the link function maps permissible mean values to $(-\infty, \infty)$.
- ▶ In practice, the canonical link function is often used by default.
- ▶ The canonical link is important for conditional inference, and it also simplifies the algorithm used to estimate the model parameters.

Fitting GLMs

- ▶ We can estimate the parameters and perform inference using maximum likelihood.
- ▶ Unlike for linear regression, the MLEs typically have to be obtained by iterative numerical methods (Newton-Raphson).
- ▶ It is also important to note that unlike for linear regression, where distribution results are exact, in general for GLMs we rely on asymptotic (large sample) results.

GLMs in R

R's `glm` function can be used to fit GLMs.

- ▶ The basic syntax (for a logistic regression) is as follows.

```
glm(y~x, family(binomial))
```

- ▶ The family option specifies which distribution in the exponential family you want to use, e.g. `gaussian`, `binomial`.
- ▶ If you do not specify the link, it will use the canonical one for the distribution you specify.
- ▶ If you want a different link, you use e.g.

```
glm(y~x, binomial(link="identity"))
```

GLMs in R - mental activity interaction model

As an example, return to the following model fitted to the morphine and mental activity trial data using the `lm` command:

Call:

```
lm(formula = mentact ~ factor(treat) * prement, data = subset(mental,
  treat < 3))
```

Residuals:

	Min	1Q	Median	3Q	Max
	-7.8281	-2.0250	-0.1053	1.8176	6.1719

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.97803	1.21662	1.626	0.11112
factor(treat)2	-1.21174	1.64558	-0.736	0.46542
prement	0.59394	0.17249	3.443	0.00127 **
factor(treat)2:prement	-0.08953	0.23348	-0.383	0.70324

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.133 on 44 degrees of freedom

Multiple R-squared: 0.3753, Adjusted R-squared: 0.3327

F-statistic: 8.81 on 3 and 44 DF, p-value: 0.0001093

GLMs in R - mental activity interaction model

Now using glm:

```
mentactInterGlm <- glm(mentact~factor(treat)*prement,  
                        data=subset(mental, treat<3),  
                        family=gaussian)
```

GLMs in R - mental activity interaction model

Call:

```
glm(formula = mentact ~ factor(treat) * prement, family = gaussian,  
     data = subset(mental, treat < 3))
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.97803	1.21662	1.626	0.11112
factor(treat)2	-1.21174	1.64558	-0.736	0.46542
prement	0.59394	0.17249	3.443	0.00127 **
factor(treat)2:prement	-0.08953	0.23348	-0.383	0.70324

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 9.81811)

Null deviance: 691.48 on 47 degrees of freedom

Residual deviance: 432.00 on 44 degrees of freedom

AIC: 251.68

Number of Fisher Scoring iterations: 2

lm vs glm

As expected, if we fit using `glm` and `family(gaussian)` we obtain identical coefficient estimates, standard errors, etc.

However, there are some differences in which/how some other quantities are displayed.

For example, using `summary` on an `lm` prints 'Residual standard error' as the estimated SD of the residuals.

Whereas `glm` reports the estimated variance of the residuals as 'Dispersion parameter for gaussian family...'.

Summary

We have

- ▶ (briefly) revised linear regression,
- ▶ introduced the exponential family, and seen that the normal and binomial distributions belong to it,
- ▶ defined the three parts of a GLM: response distribution, linear predictor, and link function,
- ▶ introduced R's `glm` function.